~~I gonna try to talk about BOχEFTagain this time, this time it's better prepared.~~ I'm going to talk about BOχEFT again; this time it's better prepared.

So in our theory there are three kinds of interactions

$$V = V^{(\mathrm{OME})} + \sum_{n=1}^{N} V_n^{(\bar{b}b)} + C(\Lambda, N) \tag{1}$$

This time I'm ~~going talk~~ going to talk about $V^{(\bar{b}b)}$ here.

It's ~~a s-channel bottomonia exchange in S-wave~~ an s-channel bottomonium exchange in the S wave at leading order LO because of S-wave or S-wave because of LO?

Answer: S wave because of LO. The BO selection rule only tells us that the compact state turns into the $\Sigma_g^{+\prime}$ configuration, and $\Sigma_g^{+\prime}$ is not a pure S wave: at the end of this talk we'll find $|\Sigma_g^{+\prime}\rangle = \frac{1}{\sqrt{3}}|^3S_1\rangle + \frac{1}{\sqrt{6}}|^3D_1\rangle - \frac{1}{\sqrt{2}}|^5D_1\rangle$, all three at the same static order. So symmetry does not remove the D waves.

What removes them is the momentum expansion. Franck–Condon makes the vertex a radial overlap with a spherical Bessel function, $\int r^2 dr\, j_L(pr) R_{1n}(r)$, and

$$j_0(pr) = 1 + \mathcal{O}\big((pr)^2\big), \qquad j_2(pr) = \frac{(pr)^2}{15} + \mathcal{O}\big((pr)^4\big),$$

so a D-wave vertex starts at $\mathcal{O}(Q^2)$, against $\mathcal{O}(Q^0)$ for the S-wave vertex [P: l. 321]. At higher orders the D-wave couplings come back and $V^{(\bar{b}b)}$ becomes a matrix in channel space [P: l. 893].

Caveat: the suppression is parametric. With $p \sim Q \approx \kappa_{X_b} = 0.695\,\mathrm{GeV}$ and the rms radii 0.41, 0.66, 0.87, 1.05 fm of the bare states $n = 1$–4, one gets $p\, r_{\mathrm{rms}} \approx 1.4, 2.3, 3.1, 3.7$ [N], which is not small. So the S-wave restriction is a counting statement, not a numerical one.

~~This is the hamiltonian of the system consists of both bottomonium and heavy mesons.~~ This is the Hamiltonian of the system, which consists of both bottomonia and heavy-meson pairs.

$$\mathcal{H} = \begin{pmatrix} H - M_{\mathrm{th}} & 0 & \cdots & \hat{g}\,\hat{g}^\dagger \\ 0 & H - M_{\mathrm{th}} & \cdots & \hat{g}\,\hat{g}^\dagger \\ \vdots & \vdots & \ddots & \vdots \\ \hat{g} & \hat{g} & \cdots & H_{\mathrm{MM}} \end{pmatrix}. \tag{2}$$

which acts on the state vector

$$|\Psi\rangle = \begin{pmatrix} \lambda_1 |\bar{b}b_1(1P)\rangle \\ \lambda_2 |\bar{b}b_1(2P)\rangle \\ \vdots \\ \phi(q)\,|B\bar{B}^*\rangle \end{pmatrix}, \qquad \sum_n |\lambda_n|^2 + \int \frac{q^2 dq}{(2\pi)^3\, 2\pi^2} |\phi(q)|^2 = 1. \tag{3}$$

Here $\phi(q) = \langle q|\Psi\rangle$, with $|q\rangle$ the meson-pair states of P_M below, normalised as $\langle q'|q\rangle = \frac{2\pi^2}{q^2}\delta(q'-q)$ [P: Eq. (30)]. The paper's Eq. (46) writes $(2\pi)^3$ instead; that is consistent only if its ket $|B\bar{B}^*\rangle$ is $\sqrt{4\pi}$ times ours, and the paper settles this 4π at l. 891 with $Y_{00} = 1/\sqrt{4\pi}$ [D].

~~To obtain the equation of heavy mesons we utilizes the completeness relation~~ To get an equation for the heavy mesons alone, we use the completeness relation

$$P_Q = \sum_n |n\rangle\langle n|, \quad P_M = \int_0^\infty \frac{q^2 dq}{2\pi^2} |q\rangle\langle q|, \quad P_Q + P_M = \mathbf{1}, \quad P_Q P_M = 0 \tag{4}$$

~~Schemecatically writing hamiltonian as~~ Schematically, we write the Hamiltonian as $\mathcal{H} = H_{QQ} + H_{MM} + \hat{g} + \hat{g}^\dagger$ where $H_{QQ} = P_Q(H - M_{\mathrm{th}})P_Q = \sum_n E_n |n\rangle\langle n|$ and $\hat{g} = P_M ~~H~~ \mathcal{H} P_Q$

now insert $\mathbf{1} = P_Q + P_M$ into the ~~schrodinger~~ Schrödinger equation $\mathcal{H}|\Psi\rangle = E|\Psi\rangle$ and project

First sandwich $\mathcal{H}$ between $\mathbf{1} = P_Q + P_M$ on both sides:

$$\mathcal{H} = (P_Q + P_M)\,\mathcal{H}\,(P_Q + P_M) = \underbrace{P_Q\mathcal{H}P_Q}_{H_{QQ}} + \underbrace{P_M\mathcal{H}P_M}_{H_{MM}} + \underbrace{P_M\mathcal{H}P_Q}_{\hat{g}} + \underbrace{P_Q\mathcal{H}P_M}_{\hat{g}^\dagger},$$

where $P_Q\mathcal{H}P_M = (P_M\mathcal{H}P_Q)^\dagger = \hat{g}^\dagger$ because $\mathcal{H}$ is Hermitian. With the space ordered as (compact, meson pair) this is the block matrix

$$\mathcal{H} = \begin{pmatrix} H_{QQ} & \hat{g}^\dagger \\ \hat{g} & H_{MM} \end{pmatrix},$$

i.e. Eq. (2) with all the bottomonium rows collected into one block.

Now write $\mathcal{H}\,|\Psi\rangle = E\,|\Psi\rangle$ as $\mathcal{H}(P_Q + P_M)\,|\Psi\rangle = E\,|\Psi\rangle$ and multiply from the left by P_Q:

$$P_Q\mathcal{H}P_Q\,|\Psi\rangle + P_Q\mathcal{H}P_M\,|\Psi\rangle = E\,P_Q\,|\Psi\rangle\,.$$

Since $P^2 = P$, we can put one more projector next to the ket: $P_Q\mathcal{H}P_Q\,|\Psi\rangle = (P_Q\mathcal{H}P_Q)(P_Q\,|\Psi\rangle) = H_{QQ}\,P_Q\,|\Psi\rangle$ and $P_Q\mathcal{H}P_M\,|\Psi\rangle = \hat{g}^\dagger\,P_M\,|\Psi\rangle$, so

$$H_{QQ}\,P_Q\,|\Psi\rangle + \hat{g}^\dagger P_M\,|\Psi\rangle = E\,P_Q\,|\Psi\rangle\,.$$

Multiplying by P_M instead gives

$$\hat{g}\,P_Q\,|\Psi\rangle + H_{MM}\,P_M\,|\Psi\rangle = E\,P_M\,|\Psi\rangle\,.$$

Collecting the $P_Q\,|\Psi\rangle$ and $P_M\,|\Psi\rangle$ terms,

one arrives at

$$(E - H_{QQ})\,P_Q\,|\Psi\rangle = \hat{g}^\dagger P_M\,|\Psi\rangle\,, \qquad (E - H_{MM})\,P_M\,|\Psi\rangle = \hat{g}\,P_Q\,|\Psi\rangle\,. \tag{5}$$

then we get $P_Q\,|\Psi\rangle = (E - H_{QQ})^{-1}\hat{g}^\dagger P_M\,|\Psi\rangle$ (for $E \neq E_n$; the inverse is taken inside the compact space).

Plug this into the second equation:

$$(E - H_{MM})\,P_M\,|\Psi\rangle = \hat{g}\,(E - H_{QQ})^{-1}\hat{g}^\dagger\,P_M\,|\Psi\rangle\,.$$

Since $H_{QQ} = \sum_n E_n\,|n\rangle\,\langle n|$ is diagonal, its inverse on the compact space is a spectral sum,

$$(E - H_{QQ})^{-1} = \sum_n \frac{|n\rangle\,\langle n|}{E - E_n},$$

which gives the second equality below.

~~so the schrodinger equation of heavy mesons and the potential appear~~ so we get a Schrödinger equation for the heavy mesons alone, and the potential appears:

$$\big[H_{MM} + \hat{V}(E)\big]P_M\,|\Psi\rangle = E\,P_M\,|\Psi\rangle\,, \qquad \hat{V}(E) = \hat{g}\,(E - H_{QQ})^{-1}\hat{g}^\dagger = \sum_n \frac{\hat{g}\,|n\rangle\,\langle n|\,\hat{g}^\dagger}{E - E_n}. \tag{6}$$

The same projectors give three more things for free.

1. **Compact amplitudes.** $\lambda_n = \langle n|\Psi\rangle = \dfrac{\langle n|\hat{g}^\dagger P_M|\Psi\rangle}{E - E_n}$: once we have the meson wave function, the bottomonium components follow.
2. **Normalisation.** For a bound state ($E = -E_B$, real) write $|\Psi_M\rangle \equiv P_M\,|\Psi\rangle$. Since $\hat{g}$ does not

depend on E and $\partial_E (E - H_{QQ})^{-1} = -(E - H_{QQ})^{-2}$,

$$\begin{aligned}\langle\Psi|\Psi\rangle &= \langle\Psi|P_M|\Psi\rangle + \langle\Psi|P_Q|\Psi\rangle = \langle\Psi_M|\Psi_M\rangle + \langle\Psi_M|\hat{g}\,(E - H_{QQ})^{-2}\hat{g}^\dagger|\Psi_M\rangle \\ &= \langle\Psi_M|\big[1 - \partial_E \hat{V}(E)\big]|\Psi_M\rangle\,.\end{aligned}$$

So the bottomonium probability [P: Eq. (F13)] is

$$\mathcal{P}_{\bar{b}b} = \langle\Psi|P_Q|\Psi\rangle = \sum_n |\lambda_n|^2 = -\langle\Psi_M|\partial_E \hat{V}|\Psi_M\rangle\,:$$

the energy dependence of $\hat{V}$ *is* the compact component.

3. **Truncation.** Split $P_Q = P_{\le N} + P_{>N}$:

$$\hat{V}(E) = \sum_{n \le N} \frac{\hat{g}\,|n\rangle\,\langle n|\,\hat{g}^\dagger}{E - E_n} + \hat{g}\,P_{>N}(E - H_{QQ})^{-1}P_{>N}\,\hat{g}^\dagger.$$

The second piece has no pole near threshold and varies slowly there, so at LO we replace it by a constant. That is the $C(\Lambda, N)$ in the first equation, and the reason it depends on N [P: Eq. (59)].

This system is non-perturbative so we need to extract information from the lattice results.

~~So the next is to see what's computed in the lattice~~ So next, let's see what is computed on the lattice.

~~What lattice sees~~ What the lattice sees: string breaking

For a static b and $\bar{b}$ at distance r, unquenched lattice QCD shows two kinds of energy levels: the colour string, whose energy rises roughly linearly with r, and a pair of static–light mesons, whose energy is nearly flat. Bulava *et al.* (2024) fit the three lowest levels with the matrix [R: Bulava 2024, Sect. 3 Eq. (6)] [P: Eq. (11)]

$$H(r) = \begin{pmatrix} V(r) & \sqrt{2}g_l & g_s \\ \sqrt{2}g_l & \hat{E}_1 & 0 \\ g_s & 0 & \hat{E}_2 \end{pmatrix}, \qquad V(r) = \hat{V}_0 + \sigma r + \frac{\gamma}{r}, \tag{7}$$

in the basis $\{\bar{b}b$ string, $B\bar{B}$, $B_s\bar{B}_s\}$.

Table 1: Lattice parameters [P: Table I], identical to Bulava 2024 Table 4 [R: Bulava 2024, Table 4]. Conversion with $\sqrt{t_0} = 0.1443(7)(13)$ fm. Used in BOχEFT: σ, γ, g_l (bold).

	$\hat{E}_1$	$\hat{E}_2$	g_l	g_s	σ	$\hat{V}_0$	γ
units of t_0	0.0034(3)	0.1005(7)	0.0256(3)	0.0134(4)	0.1061(7)	−0.835(7)	−0.434(11)
physical	5 MeV	137 MeV	**35 MeV**	18 MeV	**0.198 GeV^2**	−1.14 GeV	**−0.434**

~~So in lattice results what they observed are two energy levels, horizontal dashed lines for heavy meson energy, plus a diabatic string energy as a function of static quark separation~~ What the lattice actually measures are the energy levels, the solid lines (drawn here from the fitted model). The fit explains them with diabatic entries, the dashed lines: two flat meson-pair levels ($B\bar{B}$ and $B_s\bar{B}_s$) and a string energy that rises with the static-quark separation r.

~~and the lattice guy fit such an energy level using the hamiltonian shown above to obtain the parameters we are interested.~~ The lattice group fits these levels with the Hamiltonian above to get the parameters we are interested in.

~~now we are interested in $V(r)$ and $\sqrt{2}g_l$ which are what we need in the computing our potential.~~ For our potential we need $V(r)$ and $\sqrt{2}g_l$.

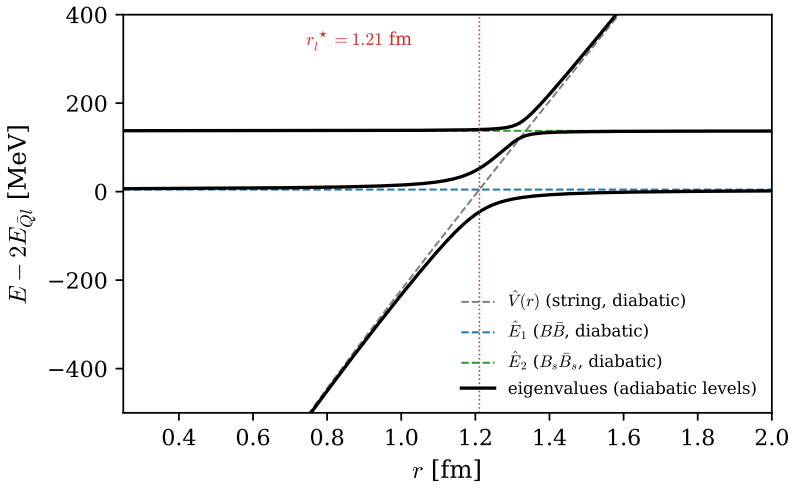
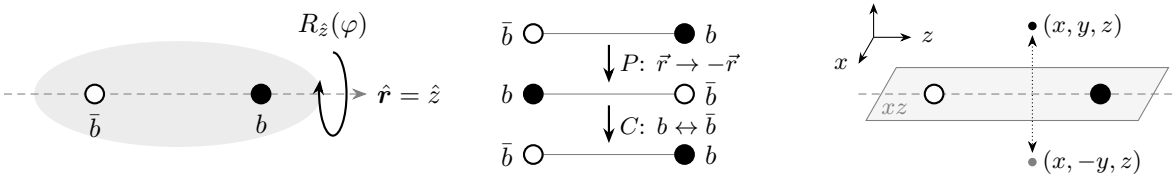


Figure 1: [N] Eigenvalues (solid) of the lattice model Eq. (7) for the physical-point parameters of Table 1, together with the diagonal ("diabatic") entries (dashed). Energies relative to $2E_{\bar{Q}l}$. The string–$B\bar{B}$ crossing is at $r_l^\star = 1.21\,\mathrm{fm}$ [R: Bulava 2024, Eq. (11)].

~~but lattice computes in $B\bar{B}$ basis, which is not ours, to relate them we need to exploit BO symmetry, which is cylindrical symmetry here~~ But the lattice works in the $B\bar{B}$ basis, which is not ours. To relate the two we need the BO symmetry, which here is a cylindrical symmetry.

~~before diving in I need to first briefly introduce the cylindrical symmetry and the so called BO quantum number Λ_η^g first.~~ Before diving in, I need to briefly introduce the cylindrical symmetry and the so-called BO quantum numbers Λ_η^σ.



(a) rotations about the axis conserve $\lambda_l = \vec{j}_l \cdot \hat{\boldsymbol{r}}$ $\Lambda = |\lambda_l|$: $\Sigma, \Pi, \Delta, \ldots$

(b) CP maps the configuration onto itself: eigenvalue $\eta = g$ $(+1)$ or u (-1)

(c) $M_{xz} = P\,R_y(\pi)$: $y \to -y$, sources stay put, $\lambda_l \to -\lambda_l$ $\Rightarrow \sigma = \pm$ only for $\Lambda = 0$

Figure 2: Symmetries of two static sources at $\pm\vec{r}/2$ on the z axis (filled: b, open: $\bar{b}$, shaded in (a): light degrees of freedom). Together they define the BO labels Λ_η^σ.

~~so angular momentum originates from SO(3) symmetry, our theory is invariant under rotations, but when there's a special axis in the space, we can assume no more when rotation involves the axis.~~ Angular momentum comes from the SO(3) rotational symmetry of our theory. Once there is a special axis in space, only rotations about that axis remain a symmetry.

~~so the SO(3) symmetry reduces to a cylindrical symmetry, and for angular momentum only it's projection to the axis $\vec{r}$ is conserved whic is $\Lambda = |\vec{L} \cdot \vec{r}|$ with $\vec{L}$ the total angular momentum of light d.o.f.~~ So SO(3) reduces to a cylindrical symmetry, and only the projection of the angular momentum onto the axis is conserved: $\lambda_l = \vec{j}_l \cdot \hat{\boldsymbol{r}}$ and $\Lambda = |\lambda_l|$, with $\vec{j}_l$ the total angular momentum of the light d.o.f. We write $\Sigma, \Pi, \Delta, \ldots$ for $\Lambda = 0, 1, 2, \ldots$ (Fig. 2a).

~~next it's $\eta = CP$ where you first use P to swap the quark position then use C to swap it's quark number~~ Next is $\eta = CP$: P swaps the positions of b and $\bar{b}$, then C exchanges quark and antiquark, which restores the configuration (Fig. 2b).

denoted by g and u for even and odd

~~and then it's $\sigma = \pm$ which is reflection according to certain plane~~ and then there is $\sigma = \pm$, the eigenvalue under reflection through a plane that contains the axis.

~~suppore we have $\vec{r}$ aligned with z axis and consider the reflection according to the xz plane~~ Suppose $\vec{r}$ is along the z axis and take the reflection through the xz plane (Fig. 2c).

we have the reflection $M_{xz} = PR_y(\pi)$: $R_y(\pi)$ sends $(x, y, z) \to (-x, y, -z)$ and P flips everything back except y, so the sources stay where they are.

and ~~it's~~ its action on a state is $PR_y(\pi) |j_l, m\rangle = P(-1)^{j_l - m} |j_l, -m\rangle$

specifically $\sigma |j_l, 0\rangle = P(-1)^{j_l} |j_l, 0\rangle$. For $\lambda_l \neq 0$ the reflection maps $\lambda_l \to -\lambda_l$, so the two states are degenerate and σ is only quoted for $\Lambda = 0$.

~~now we turn back to the lattice result, $\sqrt{2}g_l$ can be written schematically as~~ Now back to the lattice result: schematically, $\sqrt{2}g_l$ is $\langle W|H_{\text{mix}}|B\bar{B}\rangle$

where $|W\rangle$ ~~gets the quantum number~~ inherits its quantum numbers from the interpolating operator

$$\mathcal{O}_W(y, x, t) = \bar{Q}(y, t)\frac{\gamma \cdot \vec{r}}{r} W(y, x, t)Q(x, t)$$

$\gamma \cdot \vec{r}/r$ gives $|1, 0\rangle$ for the heavy spin and by HQSS we obtain

$$\sqrt{2}g_l = -\frac{1}{2} \langle W|H|1_H, 0; 1_l, 0\rangle$$

since for $B\bar{B}$ the angular momentum is coupled as $|[(s_b j_{l1})_0 (s_{\bar{b}} j_{l2})(s_{\bar{b}} j_{l2})_0]_0\rangle$

to recouple it into $|[(s_b s_{\bar{b}})_{s_H} (j_{l1} j_{l2})_{j_l}]_0\rangle$

~~because I'm lazy, I'll just copy the wigner 9j symbol~~ and the recoupling coefficient is a Wigner $9j$ symbol.

For four angular momenta the recoupling coefficient is

$$\langle [(j_1 j_2) J_{12} (j_3 j_4) J_{34}] J \,|\, [(j_1 j_3) J_{13} (j_2 j_4) J_{24}] J\rangle$$
$$= \sqrt{(2J_{12}+1)(2J_{34}+1)(2J_{13}+1)(2J_{24}+1)} \begin{Bmatrix} j_1 & j_2 & J_{12} \\ j_3 & j_4 & J_{34} \\ J_{13} & J_{24} & J \end{Bmatrix}.$$

Here $j_1 = s_b$, $j_2 = j_{l1}$, $j_3 = s_{\bar{b}}$, $j_4 = j_{l2}$ are all $\frac{1}{2}$, $J_{12} = J_{34} = 0$ because both mesons are pseudoscalars, and $J = 0$. So $J_{13} = s_H$ and $J_{24} = j_l$ must couple to zero, which means $s_H = j_l \equiv s \in \{0, 1\}$.

A $9j$ symbol with a zero reduces to a $6j$ symbol,

$$\begin{Bmatrix} a & b & e \\ c & d & e \\ f & f & 0 \end{Bmatrix} = \frac{(-1)^{b+c+e+f}}{\sqrt{(2e+1)(2f+1)}} \begin{Bmatrix} a & b & e \\ d & c & f \end{Bmatrix},$$

so for us

$$\begin{Bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ s & s & 0 \end{Bmatrix} = \frac{(-1)^{1+s}}{\sqrt{2s+1}} \begin{Bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & s \end{Bmatrix}, \qquad \begin{Bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & s \end{Bmatrix} = \frac{(-1)^{1+s}}{2},$$

where the last one is the elementary $6j$ symbol with a zero. The two signs cancel, so the $9j$ symbol is $\frac{1}{2\sqrt{2s+1}}$ and the coefficient is $\sqrt{(2s+1)^2} \cdot \frac{1}{2\sqrt{2s+1}} = \frac{\sqrt{2s+1}}{2}$:

$$|B\bar{B}\rangle = \frac{1}{2} |(0_H 0_l)_0\rangle + \frac{\sqrt{3}}{2} |(1_H 1_l)_0\rangle, \qquad \frac{1}{4} + \frac{3}{4} = 1.$$

Finally, open $(1_H 1_l)_0$ into projections on the axis,

$$|(1_H 1_l)_0\rangle = \sum_m \langle 1\, m\, 1\, -m|0\, 0\rangle \; |1_H, m\rangle \, |1_l, -m\rangle, \qquad \langle 1\, m\, 1\, -m|0\, 0\rangle = \frac{(-1)^{1-m}}{\sqrt{3}},$$

J_l	λ_l	BO quantum number
1	0	$\Sigma_g^{+\prime}$
1	±1	$\Pi_g; 1$
2	±1	$\Pi_g; 2$

so the coefficient of $|1_H, 0\rangle\,|1_l, 0\rangle$ is $\frac{\sqrt{3}}{2}\cdot\left(-\frac{1}{\sqrt{3}}\right) = -\frac{1}{2}$. Altogether

$$|B\bar{B}\rangle = \frac{1}{2}\,|0_l, 0\rangle\,|0_H, 0\rangle + \frac{1}{2}\sum_{\lambda_l=\pm1,0}(-1)^{\lambda_l+1}\,|1_l, \lambda_l\rangle\,|1_H, -\lambda_l\rangle\,,$$

which is the paper's decomposition [P: Eq. (14)].

then you see the only part with $|1_H, 0\rangle$ is $|1_H, 0; 1_l, 0\rangle$

~~that why we acquire an additional 2 in the coupling~~ that's why we pick up an extra factor of 2 in the coupling: $\langle W|H|1_H, 0; 1_l, 0\rangle = -2\sqrt{2}\,g_l$.

but this is still not what we want

we need $\langle B\bar{B}^*(^3S_1)|H|W\rangle$

~~now insert the BO basis projector we have~~ now insert the completeness relation of the BO basis, $\sum|\mathrm{BO}\rangle\langle\mathrm{BO}|$ over $\{\Sigma_g^{+\prime}, \Pi_g; 1, \Pi_g; 2\}$. The string W is Σ_g^+ and the mixing conserves the BO quantum numbers, so only $\Sigma_g^{+\prime}$ survives (BO selection rule):

$$\langle B\bar{B}^*(^3S_1)|H|W\rangle = \langle B\bar{B}^*(^3S_1)|\Sigma_g^+\,\Sigma_g^{+\prime}\rangle\,\langle\Sigma_g^+\,\Sigma_g^{+\prime}|H|W\rangle$$

and we need to show that $\langle B\bar{B}^*(^3S_1)|\Sigma_g^+\,\Sigma_g^{+\prime}\rangle = \frac{1}{3}\,\frac{1}{\sqrt{3}}$ (its square, $\frac{1}{3}$, is the S-wave probability)

so now we have the total angular momentum as $\vec{J} = \vec{s}_H + \vec{j}_l + \vec{L}$

and $J^{PC} = 1^{++}$

~~with $C = (-1)^{s_H+j_L+L}$ $P = (-1)^L$ we conclude that L is even and so for the $s_H + j_l$~~ with $C = (-1)^{s_H+j_l+L}$ and $P = (-1)^L$, $P = +$ makes L even, and then $C = +$ makes $s_H + j_l$ even,

~~since L even we must have $s_H = j_L = 1$~~ so $(s_H, j_l) = (0, 0)$ or $(1, 1)$. With $(0, 0)$ we would get $J = L$, which is even and never 1, so $s_H = j_l = 1$

and $1 \otimes 1 = 0 \oplus 1 \oplus 2$

this gives us 1^{++} meson pairs

$B\bar{B}^*(^3S_1), B\bar{B}^*(^3D_1), B^*\bar{B}^*(^5D_1)$

now we couple it in a different way $\vec{J} = \vec{J}_l + \vec{s}_H$ where $\vec{J}_l = \vec{j}_l + \vec{L}$

~~then we immediately have $J_l = 1, 2, 3$ but 3 clearly can not couple $s_H = 1$ to $J = 1$~~ then $L = 0$ gives $J_l = 1$, and $L = 2$ gives $J_l = 1, 2, 3$, but $J_l = 3$ cannot couple with $s_H = 1$ to $J = 1$

~~you will see that the combination $J_l = 2, \lambda_l = 0$ is absent, this can be seen using the CG coefficients where~~ Note that the combination $J_l = 2$, $\lambda_l = 0$ is absent. The CG coefficient shows it:

$\langle L = 2, M_L = 0; j_l = 1, M_{j_l} = 0|J_l = 2, 0\rangle = 0$

Now let's build the transformation. The two coupling orders are

$$\text{partial waves: } |L, (j_l\, s_H)S; J\rangle\,, \qquad \text{BO: } |(L\, j_l)J_l,\, s_H; J\rangle\,,$$

with $j_l = s_H = J = 1$, and we go from one to the other in three steps.

Step 1: $6j$ recoupling at fixed L [P: Eq. (20)]. Moving j_l from s_H over to L is a $6j$ symbol:

$$\langle (L\, j_l)J_l,\, s_H; J\,|\,L, (j_l\, s_H)S; J\rangle = (-1)^{L+j_l+s_H+J}\sqrt{(2J_l+1)(2S+1)}\begin{Bmatrix} L & j_l & J_l \\ s_H & J & S\end{Bmatrix}.$$

At $L=0$ nothing happens: $|(0\,1)1\rangle = |^3S_1\rangle$. At $L=2$ the phase is $(-1)^5=-1$, and

$$\begin{Bmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \end{Bmatrix} = \frac{1}{6}, \qquad \begin{Bmatrix} 2 & 1 & 1 \\ 1 & 1 & 2 \end{Bmatrix} = \begin{Bmatrix} 2 & 1 & 2 \\ 1 & 1 & 1 \end{Bmatrix} = -\frac{\sqrt{5}}{10}, \qquad \begin{Bmatrix} 2 & 1 & 2 \\ 1 & 1 & 2 \end{Bmatrix} = -\frac{1}{10},$$

so, e.g., $\langle (2\,1)1|^3D_1\rangle = -\sqrt{3\cdot 3}\cdot\frac{1}{6} = -\frac{1}{2}$. In total

$$|(2\,1)J_l{=}1\rangle = -\frac{1}{2}\,|^3D_1\rangle + \frac{\sqrt{3}}{2}\,|^5D_1\rangle\,, \qquad |(2\,1)J_l{=}2\rangle = \frac{\sqrt{3}}{2}\,|^3D_1\rangle + \frac{1}{2}\,|^5D_1\rangle\,.$$

Step 2: fix the axis [P: Eq. (21)]. States with a definite projection λ_l on $\hat{\boldsymbol{r}}$ follow from the Jacob–Wick formula

$$|J_l\,M_l;\lambda_l\rangle = \sum_L \sqrt{\frac{2L+1}{2J_l+1}}\,\langle L\,0\,j_l\,\lambda_l|J_l\,\lambda_l\rangle\;|(L\,j_l)J_l\,M_l\rangle\,.$$

The 0 is the physics: $\vec{L}=\vec{r}\times\vec{p}$ is perpendicular to $\hat{\boldsymbol{r}}$, so $\vec{L}\cdot\hat{\boldsymbol{r}}=0$ and the whole projection λ_l comes from $\vec{j}_l$. The coefficients do not depend on M_l. Case by case:

- $J_l=1$, $\lambda_l=0$: with $\langle 0\,0\,1\,0|1\,0\rangle = 1$, $\langle 1\,0\,1\,0|1\,0\rangle = 0$ and $\langle 2\,0\,1\,0|1\,0\rangle = -\sqrt{2/5}$,

$$|\Sigma_g^{+\prime}\rangle = \frac{1}{\sqrt{3}}\,|(0\,1)1\rangle - \sqrt{\frac{2}{3}}\,|(2\,1)1\rangle\,.$$

- $J_l=1$, $\lambda_l=\pm1$: $|{\pm}1\rangle = \frac{1}{\sqrt{3}}\,|(0\,1)1\rangle \mp \frac{1}{\sqrt{2}}\,|(1\,1)1\rangle + \frac{1}{\sqrt{6}}\,|(2\,1)1\rangle$. The $L=1$ piece has the wrong parity, and it is the one that flips sign between $+1$ and -1, so the parity-even state is $(|{+}1\rangle + |{-}1\rangle)/\sqrt{2}$:

$$|\Pi_g;1\rangle = \sqrt{\frac{2}{3}}\,|(0\,1)1\rangle + \frac{1}{\sqrt{3}}\,|(2\,1)1\rangle\,.$$

- $J_l=2$, $\lambda_l=\pm1$: $|{\pm}1\rangle = \sqrt{\frac{3}{10}}\,|(1\,1)2\rangle \mp \frac{1}{\sqrt{2}}\,|(2\,1)2\rangle + \frac{1}{\sqrt{5}}\,|(3\,1)2\rangle$. Now the even-$L$ piece is the one that flips sign, so the parity-even state is $(|{+}1\rangle - |{-}1\rangle)/\sqrt{2} = -\,|(2\,1)2\rangle$. Dropping the overall sign (a convention), $|\Pi_g;2\rangle = |(2\,1)2\rangle$. The paper does not spell out this sign switch.

- $J_l=2$, $\lambda_l=0$: $|0\rangle = \sqrt{\frac{2}{5}}\,|(1\,1)2\rangle - \sqrt{\frac{3}{5}}\,|(3\,1)2\rangle$ contains only odd L, i.e. negative parity. That is the full reason this state is absent: besides the vanishing $L=2$ CG, there is no even-L piece at all.

Step 3: combine [P: Eq. (22)]. Insert Step 1 into Step 2, e.g.

$$|\Sigma_g^{+\prime}\rangle = \frac{1}{\sqrt{3}}\,|^3S_1\rangle - \sqrt{\frac{2}{3}}\Big(-\frac{1}{2}\,|^3D_1\rangle + \frac{\sqrt{3}}{2}\,|^5D_1\rangle\Big) = \frac{1}{\sqrt{3}}\,|^3S_1\rangle + \frac{1}{\sqrt{6}}\,|^3D_1\rangle - \frac{1}{\sqrt{2}}\,|^5D_1\rangle\,,$$

and the same for the other two rows:

$$\begin{pmatrix} |\Sigma_g^{+\prime}\rangle \\ |\Pi_g;1\rangle \\ |\Pi_g;2\rangle \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \sqrt{\frac{2}{3}} & -\frac{1}{2\sqrt{3}} & \frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}}_{U} \begin{pmatrix} |B\bar{B}^*(^3S_1)\rangle \\ |B\bar{B}^*(^3D_1)\rangle \\ |B^*\bar{B}^*(^5D_1)\rangle \end{pmatrix}, \qquad UU^T = \mathbf{1}. \tag{8}$$

Each row is normalised ($\frac{1}{3}+\frac{1}{6}+\frac{1}{2} = \frac{2}{3}+\frac{1}{12}+\frac{1}{4} = \frac{3}{4}+\frac{1}{4} = 1$) and the rows are orthogonal: the two bases describe the same three states.

The first row is what we were after: $\langle B\bar{B}^*(^3S_1)|\Sigma_g^{+\prime}\rangle = \frac{1}{\sqrt{3}}$. Back to the projector insertion above:

$$g \equiv \langle B\bar{B}^*(^3S_1)|H|W\rangle = \underbrace{\langle B\bar{B}^*(^3S_1)|\Sigma_g^{+\prime}\rangle}_{1/\sqrt{3}}\ \underbrace{\langle\Sigma_g^{+\prime}|H|W\rangle}_{-2\sqrt{2}\,g_l},$$

$$|g| = \underbrace{\sqrt{2}}_{\text{flavour}} \times \underbrace{2}_{\text{spin}} \times \underbrace{\frac{1}{\sqrt{3}}}_{S\text{ wave}} \times g_l = \frac{2}{\sqrt{3}}\sqrt{2}\,g_l = 57\,\text{MeV}$$

for $g_l = 35\,\text{MeV}$ [P: Eq. (18)]. Here $\langle\Sigma_g^{+\prime}|H|W\rangle$ is the mixing with the normalised $\Sigma_g^{+\prime}$ configuration; by HQSS it equals $\langle 1_H, 0; 1_l, 0|H|W\rangle = -2\sqrt{2}\,g_l$ from the lattice. The sign does not matter, since $V^{(\bar{b}b)} \propto g^2$.